Photonics - Spring 2023
Review Questions

(1) A ray of red light in air is incident on the surface of an unknown non-magnetic dielectric material at an angle of 39°. It was found that the refracted ray makes an angle of 27°. Find the refractive index of this material

(2) Compare the focal length of a lens surrounded by air and the same lens surrounded by a liquid of refractive index 1.6. Explain your answer

(3) Consider a 5-mW He-Ne laser at wavelength 632.8 nm. How many photons are generated every 1 ps?

(4) Explain why an optical cavity resonator is needed in a semiconductor laser. Is it possible to have a single-mode laser? How?

(5) Discuss the main differences between edge and surface light emitting diodes. What are the best applications for each type?

(6) Consider a semiconductor Fabry-Perot optical cavity of length 200 microns with end mirrors of 80% reflectance. The refractive index of the semiconductor is 3.7. Calculate the cavity mode nearest to the free space wavelength of 1300 nm. Also, calculate the spectral width (in Hz).

(7) Calculate the thickness of a Ge crystal layer needed for absorbing 90% of incident radiation at 1550 nm. Repeat for $In_{0.53}Ga_{0.47}As$.

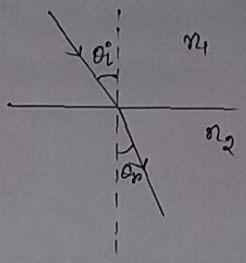
(8) A GaAs LED emits at 860 nm. It is connected to a 9 V battery and draws a current of 20 mA. If the internal efficiency of the LED is 0.85, calculate the generated photon flux

(1) A ray of red light in air is incident on the surface of an unknown non-magnetic dielectric material at an angle of 39°. It was found that the refracted ray makes an angle of 27°. Find the refractive index of this material

(1) The angle of incident of light is $\theta_i = 39°$

The angle of refraction is $\theta_r = 27°$

Here light goes from air to the dielectric material



Refractive index of air is $n_1 = 1.00$

& the refractive index of the dielectric material is n_2.

From Snell's Law

$$n_1 \sin\theta_i = n_2 \sin\theta_r$$

$$\Rightarrow n_2 = \frac{\sin\theta_i}{\sin\theta_r} n_1$$

$\theta_i = 39°$
$\theta_r = 27°$
$n_1 = 1.00$

$$\Rightarrow n_2 = \frac{\sin(39°)}{\sin(27°)} \times 1.00$$

$$\Rightarrow \boxed{n_2 = 1.386}$$

∴ The refractive index of the material is 1.386

(3) Consider a 5-mW He-Ne laser at wavelength 632.8 nm. How many photons are generated every 1 ps?

(13) (3)

$P = 5\ mW$

$\lambda = 632.8\ nm$, $t = 1\ ps$

we have

$$Pt = E = n\frac{hc}{\lambda}$$

$$\Rightarrow n = \frac{Pt\lambda}{hc}$$

$$= \frac{5\times10^{-3} \times 1\times10^{-12} \times 632.8\times10^{-9}}{6.626\times10^{-34} \times 3\times10^{8}}$$

$$\boxed{n = 1.59 \times 10^{4}} \text{ photons}$$

(2) Compare the focal length of a lens surrounded by air and the same lens surrounded by a liquid of refractive index 1.6. Explain your answer

(2) Let the radii of curvatures of the two surfaces of lens is R_1 and R_2 and f is the focal length of the lens. n_2 is the refractive index of the lens material and n_1 is the refractive index of the surroundings of the lens.

According to lens maker formula

$$\frac{1}{f} = \left(\frac{n_2}{n_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

When the lens is surrounded by air then $n_1 = 1.00$ and $n_2 = 1.5$ (for glass).

∴ The focal length of the lens is f_1

$$\frac{1}{f_1} = \left(\frac{1.5}{1.0} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\Rightarrow \frac{1}{f_1} = (0.5)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \text{ ---} (1)$$

When the length is surrounded by a liquid of refractive index 1.6. then $n_1 = 1.6$ and $n_2 = 1.5$

The focal length of the lens is f_2

$$\frac{1}{f_2} = \left(\frac{1.5}{1.6} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

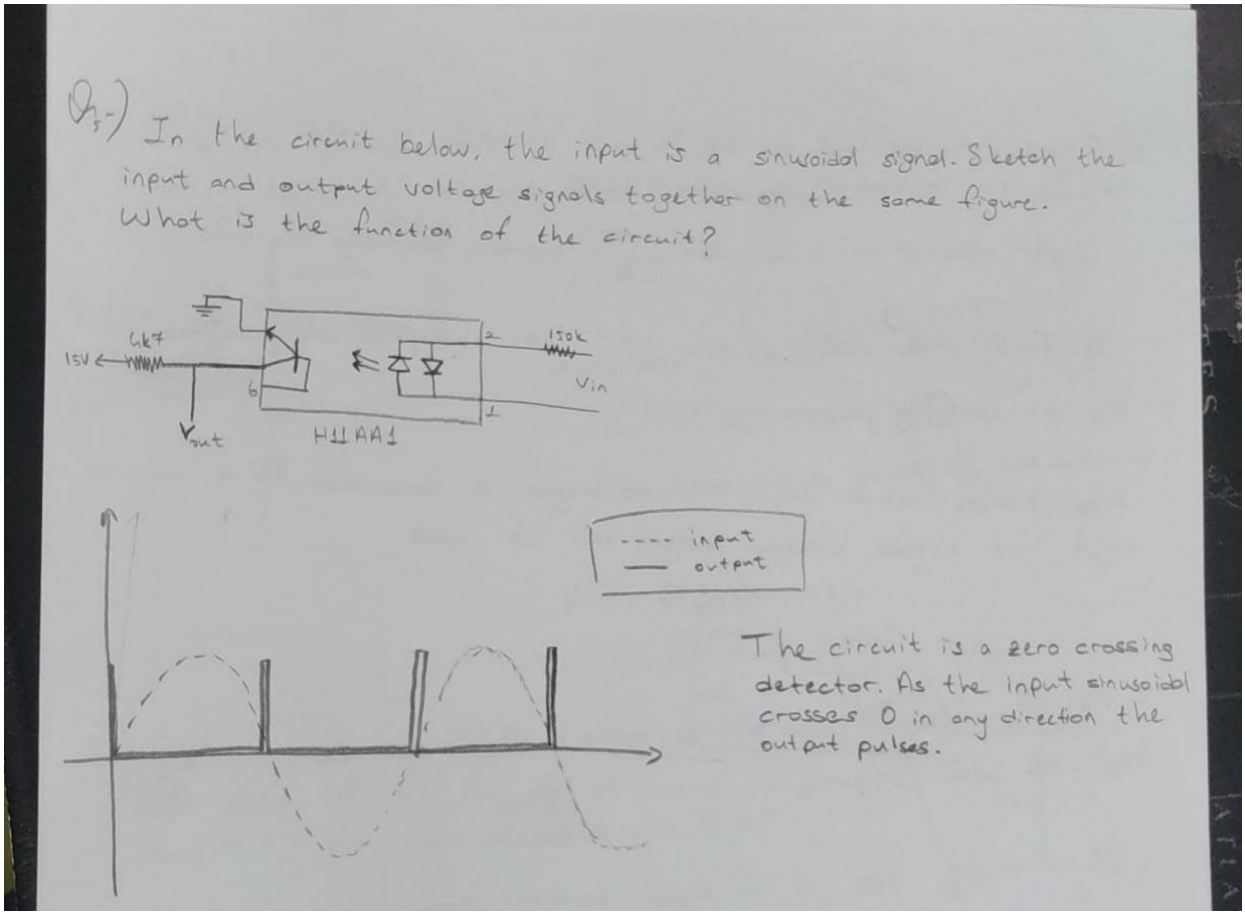
$$\Rightarrow \frac{1}{f_2} = (-0.0625)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \text{ ---} (2)$$

Dividing equation (1) by equation (2) we get

$$\frac{\frac{1}{f_1}}{\frac{1}{f_2}} = \frac{(0.5)}{(-0.0625)}$$

$$\boxed{\frac{f_2}{f_1} = -8}$$

(5) Discuss the main differences between edge and surface light emitting diodes. What are the best applications for each type?



(6) Consider a semiconductor Fabry-Perot optical cavity of length 200 microns with end mirrors of 80% reflectance. The refractive index of the semiconductor is 3.7. Calculate the cavity mode nearest to the free space wavelength of 1300 nm. Also, calculate the spectral width (in Hz).

Step 1/2

To calculate the cavity mode nearest to the free space wavelength of 1300 nm and the spectral width in Hz, we can use the following formulas:

Cavity mode :

The Fabry-Perot cavity supports resonant modes with wavelengths given by:

$$\lambda_m = 2 \times \frac{L}{m}$$

Where:
λ_m is the wavelength of the m-th mode.

L is the length of the cavity (200 microns in this case).
m is an integer representing the mode number.

We need to find the mode number m that corresponds to the free space wavelength of 1300 nm (1.3 microns).

Rearranging the formula, we have:
$\mathbf{m} = 2 \times \frac{\mathbf{L}}{\lambda_{\mathbf{m}}}$

Substituting the given values:
$\mathbf{m} = 2 \times 200\mathbf{micron}\frac{\mathbf{s}}{1.3}\mathbf{microns}$

$\mathbf{m} \approx 307.69$

Since m must be an integer, we can round it to the nearest whole number:

$\mathbf{m} \approx 308$

Therefore, the cavity mode nearest to the free space wavelength of 1300 nm is the mode with m = 308.

Spectral width :

The spectral width of the cavity modes in a Fabry-Perot cavity can be approximated using the formula:

$\Delta\lambda = \frac{\lambda^2}{2 \times \mathbf{L} \times \mathbf{n}}$

Where:

$\Delta\lambda$ **is the spectral width.**

λ **is the free space wavelength (1300 nm)** ·

L is the length of the cavity (200 microns) ·

n is the refractive index of the semiconductor (3.7) ·

Substituting the given values:

$= \Delta\lambda = \frac{(1.3\mathbf{microns})^2}{2 \times 200\mathbf{microns} \times 3.7}$

$\Delta\lambda \approx 1.38\mathbf{nm}$

To convert the spectral width from nm to Hz, we can use the formula:

$\Delta\nu = \frac{\mathbf{c}}{\lambda}$

Where:
$= \Delta\nu$ **is the spectral width in Hz** ·

c is the speed of light in vacuum (approximately 3 x 10^8 **m/s)** ·

λ **is the free space wavelength (1300 nm)** ·

Converting the wavelength to meters:
$\lambda = 1{,}300\mathbf{nm} \times 10^{-9}\frac{\mathbf{m}}{\mathbf{n}}\mathbf{m}$

$\lambda = 1.3 \times 10^{-6}\mathbf{m}$

Substituting the values:
$\Delta\nu = \frac{3\mathbf{x}10^8\frac{\mathbf{m}}{\mathbf{s}}}{1.3 \times 10^{-6}\mathbf{m}}$

$\Delta\nu \approx 2.31\mathbf{x}10^{14}\mathbf{Hz}$

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(7) Calculate the thickness of a Ge crystal layer needed for absorbing 90% of incident radiation at 1550 nm. Repeat for $In_{0.53}Ga_{0.47}As$.

Que:- Ans:- Given Ge has a incident radiation

$\alpha = 5.2 \times 10^5\ m^{-1}$ at $1.5\mu m$,

$\therefore \quad 1 - e^{(-\alpha \cdot d)} = 0.9$

Taking ln both side

$$\ln(1 - e^{(-\alpha \cdot d)} = \ln 0.9$$

$$d = \frac{1}{\alpha} \ln\left(\frac{1}{1-0.9}\right)$$

~~where d is energy gap~~

where d is thickness of Ge and In

$$d = \frac{1}{5.2 \times 10^5} \ln\left(\frac{1}{1-0.9}\right) \Rightarrow 4.428 \times 10^{-6}\ m$$

Thickness of Ge $\boxed{d = 4.428\ \mu m}$

For thickness of $In_{0.53}Ga_{0.47}$

$\alpha = 7.5 \times 10^5\ m^{-1}$ at $1.5\ \mu m$ incident radiation.

$d = \frac{1}{\alpha} \ln\left[\frac{1}{1-0.9}\right] \Rightarrow$ $d = \frac{1}{7.5 \times 10^5} \ln\left(\frac{1}{1-0.9}\right) = 3.07 \times 10^{-6} m$

$\boxed{d = 3.07\ \mu m}$

(8) A GaAs LED emits at 860 nm. It is connected to a 9 V battery and draws a current of 20 mA. If the internal efficiency of the LED is 0.85, calculate the generated photon flux

Quantum Efficiency $(\eta) = \frac{I/e}{(hf/P_{in})^{-1}} = \frac{I/e}{P_{in}/hf}$

I = current drawn

e = charge of electron

h = planck's constant

$f = \frac{c}{\lambda}$

P_{in} = input optical power

$$0.85 = \frac{I}{P_{in}} * \frac{hf}{e}$$

Given $\eta = 0.85$, $\eta = 0.85$

$I = 20mA$, $\lambda = 860nm$

$$0.85 = \frac{20\times10^{-3}}{P_{in}} \times \frac{6.625\times10^{-34}\times3\times10^{8}}{860\times10^{-9}\times1.6\times10^{-19}}$$

$$0.85 = \frac{20\times10^{-3}}{P_{in}} \times 0.01444\times10^{2}$$

$$\boxed{P_{in} = 0.03399\ W}$$

$$\boxed{P_{in} = 33.99mW}$$